

1、

2.2 求导法则

1) $u(x), v(x) \quad (u(x)+v(x))' = u'(x) + v'(x)$
 $(u_1(x) + \dots + u_n(x))' = u_1'(x) + \dots + u_n'(x)$
有限个

2) $(uv)' = u'v + uv'$

3) $\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$

4) $(\tan x)' = \sec^2 x \quad (\cot x)' = -\csc^2 x$
 $(\sec x)' = \sec x \cdot \tan x \quad (\csc x)' = -\csc x \cdot \cot x$

5) 反函数 $y=f(x) \quad x=\varphi(y) \quad \varphi'(y) = \frac{1}{f'(x)}$
eg. $y=a^x \quad x=\log_a y \quad y>0$
 $(a^x)' = \frac{1}{(\log_a y)'} = \frac{1}{\frac{1}{y \ln a}} = y \ln a = a^x \ln a$

6) $(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$
 $(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$
 $(\arctan x)' = \frac{1}{1+x^2}$
 $(\operatorname{arccot} x)' = -\frac{1}{1+x^2}$

7、复合函数的求导法则：
链式法则：

$y=f(u) \quad u=f(t) \quad t=k(w) \quad w=g(x)$
 $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dt} \cdot \frac{dt}{dw} \cdot \frac{dw}{dx}$
链式法则

eg.

复合 $\ln \sin(\cos(x^2)) \quad (uv)' = u'v + uv'$

1) $(\ln \sin x)' = \frac{1}{\sin x} \cdot \cos x \quad \text{On condition}$

2) $\left[(1+2x+3x^2)^5\right]^{\sin x} = 5(1+2x+3x^2)^4 \cdot (2+6x)$

$\left(\frac{x}{2}\sqrt{a^2-x^2}\right)' = \frac{1}{2}\sqrt{a^2-x^2} + \frac{x}{2}(\sqrt{a^2-x^2})'$
 $= \frac{1}{2}\sqrt{a^2-x^2} + \frac{x}{2} \cdot \frac{1}{\sqrt{a^2-x^2}} \cdot (-2x)$

$\left[\ln \sin(\cos(x^2))\right]'$

$$\left[\ln \sin(\cos(x^2)) \right]' = \frac{1}{\sin(\cos(x^2))} \cdot \cos(\cos(x^2)) \cdot \sin(x^2) \cdot 2x$$

$$\left[e^{\sin^2 \frac{1}{x}} \right]' = e^{\sin^2 \frac{1}{x}} \cdot 2 \sin \frac{1}{x} \cdot \cos \frac{1}{x} \cdot \left(-\frac{1}{x^2} \right)$$

$$\left(\ln \ln \ln x \right)' = \frac{1}{\ln \ln x} \cdot \frac{1}{\ln x} \cdot \frac{1}{x}$$

$$\left(\sqrt{\sin \ln \cos x} \right)' = \frac{1}{2} \cdot \frac{1}{\sqrt{\sin \ln \cos x}} \cdot \cos \ln \cos x \cdot \frac{1}{\cos x} \cdot (-\sin x)$$